PROBABILITY — MCQ ASSESSMENT

PART A — BASIC PROBABILITY

1. The probability of an event can have a value between:

A. -1 and 1
B. 0 and 1
C. 1 and 100
D. $-\infty$ and $+\infty$

2. If the probability of an event is 0, the event is:

A. Certain
B. Likely
C. Impossible
D. Equally likely

3. If the probability of an event is 1, the event is:

A. Impossible
B. Certain
C. Unlikely
D. Random

4. A fair coin is tossed once. What is the probability of getting Heads?

A. 0
B. 0.25
C. 0.5
D. 1

5. A fair six-sided die is rolled once. What is the probability of getting a 4?

A. 1/2
B. 1/4
C. 1/6
D. 4/6

6. A fair die is rolled once. What is the probability of getting an even number?

A. 1/6
B. 1/3
C. 1/2
D. 2/3

7. If P(A) = 0.3, what is P(not A)?

A. 0.3
B. 0.5
C. 0.7
D. 1.3

8. Which of the following represents the complement of event A?

A. A + B
B. $A \cap B$

C. A ∪ B
D. A'

PART B — EVENTS

9. A sample space represents:

A. Only successful outcomes
B. All possible outcomes of an experiment
C. Only failed outcomes
D. The probability of an event

10. When a die is rolled once, the sample space is:

A. {0, 1, 2, 3, 4, 5}
B. {1, 2, 3, 4, 5, 6}
C. {2, 4, 6}
D. {1, 3, 5}

11. If A represents getting an even number when rolling a die, which set represents A?

A. {1, 3, 5}
B. {2, 4, 6}
C. {1, 2, 3}
D. {4, 5, 6}

12. Two events are mutually exclusive when:

A. They always occur together
B. They cannot occur at the same time
C. They have equal probability
D. They have equal outcomes

13. Which pair of events is mutually exclusive when rolling a die?

A. Getting an even number and getting a 4
B. Getting a number greater than 3 and getting a 6
C. Getting an odd number and getting an even number
D. Getting a number less than 5 and getting a 3

PART C — ADDITION RULE

14. If two events A and B are mutually exclusive, which expression gives P(A or B)?

A. P(A) × P(B)
B. P(A) + P(B)
C. P(A) - P(B)
D. P(A) / P(B)

15. If P(A) = 0.4 and P(B) = 0.3, and A and B are mutually exclusive, what is P(A or B)?

A. 0.12
B. 0.1
C. 0.7
D. 1.2

16. If two events can occur at the same time, simply adding their probabilities may:

A. Always give the correct answer
B. Double-count their overlap
C. Make the probability zero
D. Make the events independent

17. Suppose:

P(A) = 0.5
P(B) = 0.4
P(A and B) = 0.2

What is P(A or B)?

A. 0.2
B. 0.7
C. 0.9
D. 1.1

PART D — CONDITIONAL PROBABILITY

18. Conditional probability answers which question?

A. What is the probability of A given that B has occurred?
B. What is the probability of A and B never occurring?
C. What is the probability of all possible events?
D. What is the average probability?

19. The notation P(A | B) means:

A. Probability of A and B
B. Probability of A given B
C. Probability of B given A
D. Probability of neither A nor B

20. If P(A and B) = 0.2 and P(B) = 0.5, what is P(A | B)?

A. 0.1
B. 0.2
C. 0.4
D. 0.7

21. A medical test is performed only on patients who have already been identified as high-risk. The probability of a positive result in this group is an example of:

A. Conditional probability
B. Independent probability
C. Complementary probability
D. Uniform probability

22. If knowing that event B occurred changes the probability of event A, then A and B are generally:

A. Independent
B. Dependent
C. Mutually exclusive
D. Identical

PART E — INDEPENDENT EVENTS

23. Two events are independent when:

A. They cannot occur together
B. The occurrence of one does not affect the probability of the other
C. They always occur together
D. Their probabilities are equal

24. A fair coin is tossed twice. Is the result of the second toss affected by the result of the first toss?

A. Yes, always
B. No
C. Only when the first toss is Heads
D. Only when the first toss is Tails

25. If A and B are independent, which expression gives P(A and B)?

A. P(A) + P(B)
B. P(A) - P(B)
C. P(A) × P(B)
D. P(A) / P(B)

26. If P(A) = 0.5 and P(B) = 0.2, and A and B are independent, what is P(A and B)?

A. 0.1
B. 0.3
C. 0.7
D. 1.0

27. Which pair of events is most likely to be independent?

A. Drawing two cards from a deck without replacement
B. Tossing a coin twice
C. Selecting two students without replacement
D. Taking two balls from a bag without replacement

PART F — COUNTING AND COMBINATORICS

28. A fair die is rolled twice. How many possible ordered outcomes are there?

A. 6
B. 12
C. 36
D. 42

29. A coin is tossed three times. How many possible outcomes are there?

A. 3
B. 6
C. 8
D. 9

30. If there are 4 shirts and 3 pairs of trousers, how many different shirt-trouser combinations are possible?

A. 7
B. 12
C. 16
D. 24

31. Which concept is used when the order of selection matters?

A. Combination
B. Permutation
C. Mean
D. Variance

32. Which concept is used when the order of selection does NOT matter?

A. Permutation
B. Combination
C. Conditional probability
D. Independence

33. In how many ways can 3 different students be arranged in a row?

A. 3
B. 6
C. 9
D. 12

PART G — RANDOM VARIABLES

34. A random variable is:

A. A variable whose value is determined by the outcome of a random experiment
B. A variable that must always be random in a computer program
C. A variable that can only contain integers
D. A variable with probability 1

35. Which of the following is an example of a discrete random variable?

A. Height of a person
B. Temperature
C. Number of defective products
D. Weight of a person

36. Which of the following is an example of a continuous random variable?

A. Number of students
B. Number of defective items
C. Number of cars
D. Temperature

37. The sum of probabilities of all possible outcomes of a discrete random variable should be:

A. 0
B. 0.5
C. 1
D. Greater than 1

PART H — EXPECTED VALUE

38. Expected value can be interpreted as:

A. The most frequent outcome in every experiment
B. The long-run weighted average outcome
C. The maximum possible outcome
D. The minimum possible outcome

39. A random variable X has:

X = 0 with probability 0.5
X = 10 with probability 0.5

What is its expected value?

A. 2.5
B. 5
C. 10
D. 20

40. A game gives ₹100 with probability 0.2 and ₹0 with probability 0.8. What is the expected value?

A. ₹10
B. ₹20
C. ₹50
D. ₹80

PART I — BAYESIAN THINKING

41. Bayes' theorem is primarily concerned with:

A. Updating probabilities based on new evidence
B. Calculating averages
C. Finding matrix rank
D. Calculating variance

42. Suppose a medical test is positive. You want to know the probability that the patient actually has the disease. Which type of probability is most relevant?

A. P(Test Positive)
B. P(Disease)
C. P(Disease | Test Positive)
D. P(Test Positive | No Disease)

43. Why can a highly accurate medical test still produce many false positives?

A. The disease may be very common
B. The disease may be very rare, affecting the base rate
C. Probability cannot be used for medical testing
D. Accuracy and probability are always unrelated

44. In Bayesian reasoning, prior probability represents:

A. Probability after observing evidence
B. Initial belief before considering new evidence
C. Probability of an impossible event
D. Probability of two independent events

PART J — APPLICATION-BASED QUESTIONS

45. A fraud detection system flags 5% of all transactions. If a transaction is randomly selected, what does 5% represent?

A. Conditional probability
B. Prior probability
C. Expected value
D. Variance

46. A dataset contains 1,000 transactions, of which 50 are fraudulent. What is the probability that a randomly selected transaction is fraudulent?

A. 0.005
B. 0.05
C. 0.5
D. 5

47. A spam filter identifies an email as spam. You want to know the probability that it actually is spam. Which probability is most relevant?

A. P(Spam)
B. P(Flagged)
C. P(Spam | Flagged)
D. P(Flagged | Not Spam)

48. A classifier predicts that 90% of emails are legitimate. Does this necessarily mean that 90% of its predictions are correct?

A. Yes, always
B. No
C. Only if the dataset is large
D. Only if the classes are balanced

49. If an event has probability 0.8, which statement is correct?

A. It will occur exactly 80% of the time in every small sample
B. It is guaranteed to occur
C. Over many repetitions, its relative frequency is expected to approach 0.8
D. It cannot fail

50. A model predicts whether a patient has a disease. The disease occurs in only 1% of the population. Why is the 1% figure important when interpreting a positive prediction?

A. It represents the base rate
B. It represents the model accuracy
C. It represents the model's variance
D. It represents the sample size